

Reinforced Diffusion on Graphs for Decentralized Multi-Robot Navigation with Smooth Formation Adaptation

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Abstract— Coordinated multi-robot navigation in unstructured environments often requires maintaining formations, but rigid formations in constrained spaces necessitate adaptive formation control. Diffusion models generate smooth navigation trajectories through probabilistic denoising, but rely on expert demonstrations that are difficult to obtain for multi-robot teams. Online Reinforcement Learning (RL) avoids demonstrations by learning through environment interaction, but its discrete decision-making often produces jerky motions. In multi-robot settings, centralized diffusion with online RL becomes intractable as the joint action space grows exponentially with team size. To overcome these limitations, we propose *Reinforced Diffusion on Graphs* (ReDiG), one of the first principled multi-robot learning approaches that enables fully decentralized coordinated navigation with smooth formation adaptation. ReDiG uses a graph network to aggregate neighbor information via message passing and produce a team-level representation that conditions each robot’s diffusion policy for team-aware, smooth trajectory generation. ReDiG further formulates graph-conditioned diffusion policy refinement as an online actor-critic RL problem to adapt diffusion policies under dynamic teaming and environmental feedback, while using a spring-damper reward to encourage synchronized formation adaptation. ReDiG is trained in an end-to-end unified manner, with the diffusion and critic gradients jointly backpropagated through their networks and the graph network. Extensive experiments with both simulated and physical robot teams demonstrate that ReDiG achieves state-of-the-art coordinated navigation while enabling smooth and adaptive formation control.

I. INTRODUCTION

Multi-robot systems have received sustained attention due to their intrinsic advantages in redundancy [1], operational efficiency [2], and scalability [3], which render multi-robot systems well suited for large-scale, real-world applications, including search and rescue [4], [5], cooperative transportation [6], [7], and space exploration [8], [9].

Coordinated multi-robot navigation is an essential capability that enables teams to traverse environments in a synchronized manner. Coordinated navigation usually requires maintaining specific formations, e.g., wedge formations to enhance area coverage and visibility. However, as shown in Figure 1, rigid formations hinder progress in constrained spaces, such as narrow bridges. These scenarios necessitate *formation adaptation*, allowing robots to adjust their relative positions in space and time while preserving coordination. Effective adaptation also requires *smooth trajectories*, since frequent motion corrections for formation maintenance can reduce efficiency and stability. Crucially, smooth and adaptive coordination must be achieved under *decentralized control*, where robots communicate locally to ensure consistent, scalable coordination across the team.

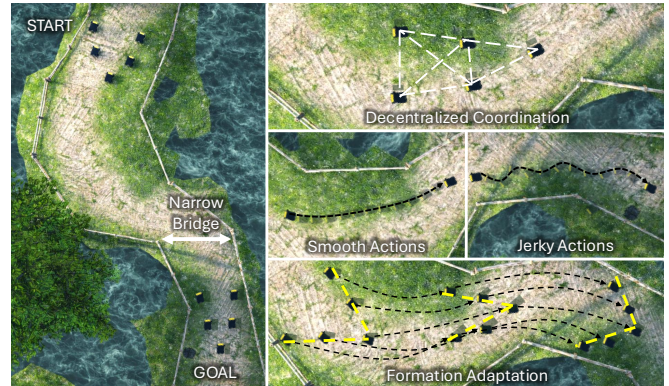


Fig. 1. When a robot team in a wedge formation approaches a narrow bridge, the robots must adapt the formation to safely traverse the bridge while moving smoothly toward their goals and coordinating in a decentralized manner.

Classic formation control strategies, such as leader-follower methods [10], rely on rigid predefined formations without adaptation. Optimization-based methods, such as Model Predictive Control [11], often trade off collision avoidance and precise formation maintenance, with high computational costs for long-horizon nonlinear optimization. Diffusion models generate smooth trajectories through iterative denoising [12] and have been applied to centralized multi-robot control [13]. However, they rely on expert demonstrations, which are difficult to obtain for synchronization across multiple robots. Online reinforcement learning (RL) enables robots to learn through environment interaction [14], but its discrete decisions often produce jerky trajectories and unstable convergence in decentralized multi-robot systems due to non-stationary interactions. Recent methods integrate diffusion models with online RL to refine policies and reduce reliance on expert demonstrations, but have primarily focused on single-robot control [15], [16]. Extending this paradigm to multi-robot systems remains challenging. First, the joint action space grows exponentially with the number of robots, making centralized policy learning intractable. Second, unconditioned diffusion policies lack mechanisms to encode team context and inter-robot dependencies for coordinated behavior.

To address these limitations and enable decentralized multi-robot navigation with smooth formation adaptation, we propose *Reinforced Diffusion on Graphs* (ReDiG). ReDiG models a robot team as a graph, where nodes represent robots and their attributes, and edges encode spatial relationships within a communication radius. A graph neural network aggregates

information from neighboring robots via message passing to compute a team-level embedding. This embedding conditions each robot’s diffusion policy, enabling team-aware action generation with smooth trajectories. ReDiG formulates the graph-conditioned diffusion policy refinement as an online actor-critic RL problem, enabling adaptation to dynamic team configurations and environmental feedback. A spring-damper reward is incorporated into the RL-based policy refinement for synchronized adaptation. ReDiG, comprising the graph network, diffusion policy, and online critic, is trained in an end-to-end unified manner, with diffusion and critic gradients jointly backpropagated through the policy and graph networks.

Our primary contribution is ReDiG, a principled multi-robot learning framework for fully decentralized coordinated navigation with smooth formation adaptation. ReDiG includes two key novelties:

- **Decentralized graph-conditioned diffusion policy** that avoids the unscalable centralized joint action space by aggregating neighbor information via message passing and explicitly encoding team awareness through graph-based conditioning to generate smooth trajectories via iterative denoising.
- **Online RL-based diffusion policy refinement** that adapts graph-conditioned diffusion policies to dynamic teaming and environmental feedback, reducing reliance on idealized and hard-to-obtain expert demonstrations while encouraging synchronized formation adaptation via a spring-damper reward.

II. RELATED WORK

A. Coordinated Multi-Robot Navigation

Traditional formation control approaches include leader-follower methods [10], [11], where followers track leaders, and virtual-region methods [17], [18], which predefine geometric regions. Model Predictive Control [19] maintains formations via constrained optimization, incurring high costs for long-horizon nonlinear planning. However, they often rely on rigid formations without adaptation, requiring precise shape maintenance in constrained spaces.

For learning-based methods, GNNs have been widely used to represent multi-robot systems [20], [21]. RL has been widely applied in multi-robot navigation [14], [22], enabling robot teams to learn complex, coordinated behaviors that are difficult to design manually. However, existing RL methods have three key limitations. First, they fail to effectively encode the spatial structure needed for formation adaptation. Second, they rely on step-wise decision-making [23], causing robots to frequently adjust their motion to maximize immediate rewards, which leads to jerky trajectories and reduced motion smoothness. Third, they usually exhibit slow or unstable convergence due to non-stationary multi-agent interactions.

Diffusion models have gained attention in robotics for generating smooth trajectories through iterative denoising. In single-robot settings, they sample plans conditioned on task objectives [24] and environmental context [25], using hierarchical structures to address long-horizon planning [12].

Recent works have extended diffusion models to multi-robot systems. These methods generate trajectories and incorporate guidance from cost functions [26], constraints [27], [28], and dynamic models [29] to improve feasibility. However, applying diffusion models to multi-robot systems remains challenging due to the need for large-scale, high-quality expert demonstrations, which are hard to obtain. Moreover, existing approaches typically operate under centralized control, limiting scalability and practicality in real-world deployments.

B. Integrating Diffusion Models with RL

Diffusion for Offline RL. Diffusion models have been combined with offline RL to improve policy using reward signals. Prior work enhances generalization through representation reconstruction and distributional factorization [30], while improving stability via consistency modeling [31]. Representative methods include trajectory-level reward guidance [32], conditional diffusion [33], and transformer diffusion [34]. Diffusion-based offline learning has also been extended to multi-robot settings [35]. However, for complex behaviors that demand coordination and synchronization, which are rarely available in offline datasets, offline RL struggles to learn behaviors that are not present in expert demonstrations.

Diffusion for Online RL. Diffusion-based online RL enables interaction-driven exploration beyond expert demonstrations. DIPO [15] integrates diffusion policies into online RL for policy improvement, and uses a critic to refine actions through gradient-ascent updates to obtain higher rewards. CPQL [36] shows that one-step consistency models can serve as policies in online RL, achieving a strong balance between exploration and exploitation. QSM [37] aligns the score function with the gradient of a Q-function, connecting the denoising process to action-value learning. QVPO [16] introduces a Q-weighted variational loss to ensure robust policy improvement by approximating the policy objective. However, these methods have not been extended to multi-robot systems and face two limitations. First, modeling and optimizing the joint action distribution becomes increasingly intractable as the team size grows due to the rapidly expanding action space. Second, unconditioned diffusion samples actions without team context, providing no explicit mechanism to encode inter-robot dependencies and thus limiting coordinated action generation required for formation control.

III. PRELIMINARIES

Diffusion Policy. Diffusion models [38] define a probabilistic generative process consisting of a forward (noising) process and a reverse (denoising) process. In the forward process, Gaussian noise is progressively added to the ground-truth action a_i^0 according to a variance schedule $\{\beta^k\}_{k=1}^K$ with $\beta^k \in (0, 1)$, which results in a sequence of noisy actions $a^1, \dots, a^K$. The forward process at step k is given by:

$$q(a^k | a^{k-1}) = \mathcal{N}(a^k; \sqrt{1 - \beta^k} a^{k-1}, \beta^k \mathbf{I})$$

In the reverse process, the diffusion model iteratively denoises a^K to recover a^0 . At each step k , the diffusion model predicts a^{k-1} from a^k , progressively refining the actions toward the

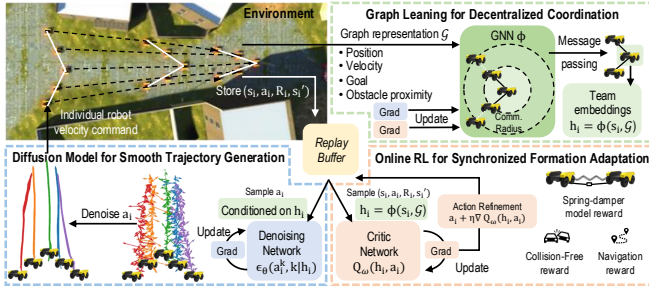


Fig. 2. ReDiG enables decentralized coordinated multi-robot navigation with smooth and adaptive formation control. Under end-to-end training, ReDiG unifies graph-based decentralized coordination, diffusion models for smooth trajectory generation, and online RL for graph-conditioned diffusion policy refinement, enabling synchronized formation adaptation.

ground truth. The reverse process is controlled by three coefficients. First, $\lambda^k = 1/\sqrt{1 - \beta^k}$ scales the denoised prediction at step k . Second, $\alpha^k = \prod_{i=1}^k (1 - \beta^i)$ accumulates the forward noise schedule, progressively reducing the influence of noise. Third, σ^k , derived from β^k , controls exploration by regulating the magnitude of noise injected during sampling.

Online RL. An agent interacts with an environment modeled as a Markov Decision Process (MDP), $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(s'|s, a)$ is the (unknown) transition dynamics, $r(s, a)$ is the reward, and $\gamma \in [0, 1]$ is a discount factor. At each step t , the agent observes the current state $s_t \in \mathcal{S}$, selects an action $a_t \sim \pi_t(\cdot|s_t)$ according to its policy π_t , receives a reward $r_t = r(s_t, a_t)$, and transitions to the next state $s_{t+1} \sim P(\cdot|s_t, a_t)$. In online RL [39], the policy π_t is updated incrementally based upon observed history (s_t, a_t, r_t, s_{t+1}) , with the goal of maximizing the expected discounted return:

$$J(\pi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid a_t \sim \pi(\cdot | s_t) \right]$$

Online RL differs from offline RL in that the agent collects and updates experience sequentially, rather than learning from a fixed dataset. In robotics, this enables continuous adaptation to changing dynamics and previously unseen environments.

IV. APPROACH

We introduce a novel unified learning-based approach called ReDiG (*Reinforced Diffusion on Graphs*), as shown in Figure 2. Our goal is to address decentralized multi-robot navigation with adaptive formation control and smooth robot trajectories. We aim to enable three multi-robot capabilities:

- **Decentralized Coordination:** Each robot coordinates with its teammates via message passing, achieving multi-robot navigation in a fully decentralized manner.
- **Smooth Trajectory Generation:** Each robot generates smooth, collision-free trajectories that ensure non-jerky movements and efficient goal reaching.
- **Synchronized Formation Adaptation:** Robots synchronize their movements in both time and space to maintain formations while dynamically adapting to constrained environments.

A. Decentralized Team-Aware Generation of Smooth Actions

We represent a team of n robots as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$, where the node set $\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ denotes the robots and $\mathcal{A} \in \{0, 1\}^{n \times n}$ is the adjacency matrix encoding local interactions. An edge exists between robots i and j if their distance is within a prescribed interaction radius $r > 0$, i.e., $a_{ij} = \mathbb{I}(\|\mathbf{p}_i - \mathbf{p}_j\|_2 \leq r)$, where $\mathbb{I}(\cdot)$ is the indicator function. Each robot $i \in \mathcal{V}$ has continuous-valued attributes $\mathbf{v}_i = (\mathbf{p}_i, \mathbf{q}_i, \mathbf{g}_i)$, where $\mathbf{p}_i = (p_i^x, p_i^y) \in \mathbb{R}^2$ denotes its planar position, $\mathbf{q}_i = (q_i^x, q_i^y) \in \mathbb{R}^2$ is its observed velocity, and $\mathbf{g}_i = (g_i^x, g_i^y) \in \mathbb{R}^2$ is its goal location. We define the state of robot i as $\mathbf{s}_i = (\mathbf{p}_i, \mathbf{q}_i, \mathbf{g}_i, d_i)$, where $d_i \in \mathbb{R}_{\geq 0}$ denotes the Euclidean distance from robot i to the nearest environment obstacle. Assuming velocity-controlled dynamics, the action of robot i is defined as $\mathbf{a}_i = (v_i^x, v_i^y) \in \mathbb{R}^2$, where v_i^x and v_i^y denote the commanded velocities along the x - and y -axes, respectively.

Given $\mathcal{G}$ and the state $\mathbf{s}_i$ of robot i , we use a graph network ϕ to encode local spatial interactions and compute a team-state embedding $\mathbf{h}_i = \phi(\mathbf{s}_i, \mathcal{G})$ for each robot. ϕ consists of linear layers that map $\mathbf{s}_i$ to an individual latent embedding $\mathbf{z}_i$ via $\mathbf{z}_i = \mathbf{W}^z \mathbf{s}_i$, where $\mathbf{W}^z$ is a learnable weight matrix. The team-state embedding $\mathbf{h}_i$ is obtained through local message passing by aggregating neighboring information as $\mathbf{h}_i = \mathbf{W}^h \mathbf{z}_i + \sum_{j \in \mathcal{N}(i)} \mathbf{W}^h (\mathbf{z}_j - \mathbf{z}_i)$, where $\mathbf{W}^h$ is a learnable matrix that captures the relative importance of neighboring information during message passing, and $\mathcal{N}(i)$ denotes the neighbors of robot i induced by $\mathcal{G}$. The first term encodes the individual state of robot i , while the second term captures spatial relationships with neighboring robots, thereby encoding team-level context. As aggregation relies exclusively on locally exchanged messages within $\mathcal{N}(i)$, ϕ enables scalable, *fully decentralized* information sharing and control for robot teams.

To generate team-aware actions $\mathbf{a}_i$, we can sample $\mathbf{a}_i$ from the true conditional action distribution $\pi^*(\mathbf{a}_i|\mathbf{h}_i)$. However, directly sampling from it is intractable due to the high dimensionality of the continuous action space. To address this, we propose a graph-conditioned diffusion policy $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$, parameterized by θ , which conditions on the team embedding $\mathbf{h}_i$ to approximate $\pi^*(\mathbf{a}_i|\mathbf{h}_i)$ through a parameterized denoising process. $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$ uses a neural network $\epsilon_\theta(\mathbf{a}_i^k, k|\mathbf{h}_i)$ to estimate the injected noise, and the reverse denoising process at each step k is defined as

$$\mathbf{a}_i^{k-1} = \lambda^k (\mathbf{a}_i^k - \alpha^k \epsilon_\theta(\mathbf{a}_i^k, k | \mathbf{h}_i)) + \sigma^k \mathbf{z}, \quad (1)$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ is standard Gaussian. $\epsilon_\theta(\mathbf{a}_i^k, k)$ produces team-aware actions in a coarse-to-fine fashion, sequentially reconstructing low-frequency motion components before refining high-frequency details at each denoising step. By estimating the conditional expectation of the noise given the corrupted actions, statistically inconsistent high-frequency components are averaged out, thereby enabling smooth trajectory generation. Let $\mathbf{a}_i^k = \sqrt{\bar{\alpha}_k} \mathbf{a}_i^0 + \sqrt{1 - \bar{\alpha}_k} \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. The loss function to learn $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$ is defined as

$$\min_{\theta} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i^0, \epsilon, k} \left[\|\epsilon - \epsilon_\theta(\mathbf{a}_i^k, k | \mathbf{h}_i)\|_2^2 \right]. \quad (2)$$

B. Online RL Refinement of $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$ for Synchronized Formation Adaptation

The graph-conditioned diffusion policy $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$ is trained on demonstrations $\{\mathbf{a}_i^0\}$, limiting its ability to generate actions beyond the demonstrated set. In multi-robot systems, obtaining high-quality demonstrations is challenging due to the need for coordinated, synchronized control across multiple robots.

We introduce ReDiG, a new approach for refining diffusion policies via online RL over graph representations of multi-robot systems. By formulating diffusion policy refinement as an online actor-critic RL problem, ReDiG enables diffusion policies to adapt to dynamic team configurations and environmental feedback, thereby reducing reliance on idealized and hard-to-obtain expert demonstrations.

Diffusion Policy Refinement via Online RL. We define the graph-conditioned diffusion policy $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$ as a stochastic generative actor. Action samples $\mathbf{a}_i \sim \pi_\theta(\cdot|\mathbf{h}_i)$ are iteratively refined using value-based feedback from a critic $Q_\omega(\mathbf{h}_i, \mathbf{a}_i)$ via gradient ascent on the critic value,

$$\mathbf{a}_i \leftarrow \mathbf{a}_i + \eta \nabla_{\mathbf{a}_i} Q_\omega(\mathbf{h}_i, \mathbf{a}_i), \quad (3)$$

where $\eta > 0$ denotes the step size controlling the magnitude of the action update, and $\nabla_{\mathbf{a}_i} Q_\omega(\mathbf{h}_i, \mathbf{a}_i)$ denotes the gradient of Q_ω with respect to $\mathbf{a}_i$, specifying the refinement direction that increases the critic value.

The refined action $\mathbf{a}_i$ is inserted into the replay buffer $\mathcal{M}$ and used as an improved supervision sample $\mathbf{a}_i^0$ for optimizing $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$. Therefore, ReDiG induces an online off-policy RL paradigm, where value-based feedback is employed during training to refine $\mathbf{a}_i$, and the learned $\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)$ directly generates actions $\mathbf{a}_i$ during execution, without reliance on critic evaluation. Parameterized as a multilayer perceptron (MLP), the critic $Q_\omega(\mathbf{h}_i, \mathbf{a}_i)$ approximates the value function defined over the joint space $(\mathbf{h}_i, \mathbf{a}_i)$. $Q_\omega(\mathbf{h}_i, \mathbf{a}_i)$ is trained by minimizing the mean-squared Bellman error $\mathbb{E}[(Q_\omega(\mathbf{h}_i, \mathbf{a}_i) - y_i)^2]$, and the target is defined as $y_i = R(\mathbf{s}_i, \mathbf{a}_i) + \gamma Q_\omega(\mathbf{h}'_i, \mathbf{a}'_i)$. To stabilize training and mitigate overestimation bias, we employ a pair of independently parameterized critic networks, Q_{ω_1} and Q_{ω_2} . The resulting critic loss is defined by

$$\mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} \left[(Q_{\omega_1}(\mathbf{h}_i, \mathbf{a}_i) - y_i)^2 + (Q_{\omega_2}(\mathbf{h}_i, \mathbf{a}_i) - y_i)^2 \right]. \quad (4)$$

During action refinement, we compute action gradients from the conservative critic estimate $\min\{Q_{\omega_1}, Q_{\omega_2}\}$ to reduce overestimation and improve the stability of policy updates.

Synchronized Formation Adaptation. To maintain and adapt formations in constrained environments, robot teams must coordinate individual motions to preserve spatial configurations over time. We design a reward function inspired by spring-damper models [40], [41], enabling synchronized formation adaptation within the online RL paradigm. The spring component penalizes inter-robot distance deviations, encouraging compact formations in cluttered environments. For robot pair (i, j) , the spatial deviation is $|d_{i,j} - p_{i,j}|$, where $d_{i,j}$ denotes the adaptive target distance (e.g., scaled according to environmental constraints) and $p_{i,j} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$ is the distance. The damper component ensures temporal consistency

Algorithm 1: Unified Training for ReDiG

Input : Team graph $\mathcal{G}$, robot states $\{\mathbf{s}_i\}$, replay buffer $\mathcal{M}$
Output : Trained graph network ϕ , denoising network ϵ_θ , and critic networks $Q_{\omega_1}, Q_{\omega_2}$

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1 Initialize  $\phi, \epsilon_\theta, Q_{\omega_1}, Q_{\omega_2}$ ;
2 while Not converged do
3   Compute team-level embeddings  $\mathbf{h}_i = \phi(\mathbf{s}_i, \mathcal{G})$ ;
4   Sample actions  $\{\mathbf{a}_i\}^n \sim \{\pi_\theta(\mathbf{a}_i|\mathbf{h}_i)\}^n$ ;
5   Compute  $\mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i^0} [\|\epsilon_k - \epsilon_\theta(\mathbf{a}_i^0 + \epsilon_k, k|\mathbf{h}_i)\|_2^2]$ ;
6   Update  $\epsilon_\theta$  based on  $\nabla_{\theta} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i^0} [\cdot]$  acc. to Eq. (2);
7   Execute  $\{\mathbf{a}_i\}^n$  and insert transition  $(\mathbf{s}_i, \mathbf{a}_i, R_i, \mathbf{s}'_i)$  to  $\mathcal{M}$ ;
8   Sample  $(\mathbf{s}_i, \mathbf{a}_i, R_i, \mathbf{s}'_i)$  from  $\mathcal{M}$ ;
9   Compute
       $\mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [(Q_{\omega_1}(\mathbf{h}_i, \mathbf{a}_i) - y_i)^2 + (Q_{\omega_2}(\mathbf{h}_i, \mathbf{a}_i) - y_i)^2]$ ;
10  Refine action  $\mathbf{a}_i = \mathbf{a}_i + \eta \nabla_{\mathbf{a}_i} Q_\omega(\mathbf{h}_i, \mathbf{a}_i)$  and insert
      updated transition  $(\mathbf{s}_i, \mathbf{a}_i, R_i, \mathbf{s}'_i)$  to  $\mathcal{M}$ ;
11  Update  $Q_{\omega_1}$  and  $Q_{\omega_2}$  based on  $\nabla_{\omega_1} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$  and
       $\nabla_{\omega_2} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$  acc. to Eq.(4);
12  Update  $\phi$  based on  $\nabla_{\theta} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i^0} [\cdot]$ ,  $\nabla_{\omega_1} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$ , and
       $\nabla_{\omega_2} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$ ;
13 end
14 return  $\phi, \epsilon_\theta, Q_{\omega_1}, Q_{\omega_2}$ 

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by regulating relative velocities as $q_{i,j} = \|\mathbf{q}_i - \mathbf{q}_j\|_2$, which suppresses oscillations. The resulting spring-damper reward is defined as $R^{adp} = \sum_{\mathbf{v}_i, \mathbf{v}_j \in \mathcal{V}} -\rho |d_{i,j} - p_{i,j}| - (1 - \rho) q_{i,j}$, where $\rho \in [0, 1]$ balances the spring and damper components. This reward promotes team-level synchronization by enforcing spatial and temporal coordination, leading to smooth, compact, and stable multi-robot formation adaptation.

C. Unified Training and Decentralized Execution

Unified Training. Algorithm 1 presents the unified training procedure for ReDiG. We compute gradients $\nabla_{\theta} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i^0} [\cdot]$ from the diffusion loss in Eq. (2), and gradients $\nabla_{\omega_1} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$ and $\nabla_{\omega_2} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$ from the critic loss in Eq. (4). Then, $\nabla_{\theta} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i^0} [\cdot]$ is backpropagated to update the denoising network ϵ_θ . The gradients of the critic loss $\nabla_{\omega_1} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$ and $\nabla_{\omega_2} \mathbb{E}_{\mathbf{h}_i, \mathbf{a}_i} [\cdot]$ are used to update the critics Q_{ω_1} and Q_{ω_2} . Finally, these gradients are further propagated through the graph network ϕ to update its parameters $\mathbf{W}^z$ and $\mathbf{W}^h$. The per-iteration training time complexity is dominated by $\mathcal{O}(n^2)$, where n is the number of robots (details are provided in the supplementary material).

Decentralized Execution. ReDiG executes fully decentralized closed-loop multi-robot control. Each robot maintains a local occupancy map and exchanges state information with nearby teammates via message passing within a communication radius (e.g., over Wi-Fi). During decentralized execution, each robot i independently applies the trained graph network ϕ , with shared parameters $\mathbf{W}^z$ and $\mathbf{W}^h$, to aggregate information from its neighbors and compute its local embedding $\mathbf{h}_i$. Each robot i then executes its own instance of the trained graph-conditioned diffusion policy $\pi_\theta(\cdot|\mathbf{h}_i)$ to generate the velocity commands $\mathbf{a}_i$, which enables fully decentralized control of the multi-robot system in a closed-loop manner. The execution time complexity is dominated by $\mathcal{O}(n + T_d)$, where T_d is the number of diffusion steps.

TABLE I
QUANTITATIVE COMPARISON OF REDiG WITH PRIOR AND BASELINE METHODS FROM GAZEBO SIMULATIONS IN ROS2.

Method	Circle Formation					Wedge Formation					Line Formation				
	SR (%)	TT (sec)	$\delta < 0.5$	$\delta < 0.1$	$\delta < 0.03$	SR (%)	TT (sec)	$\delta < 0.5$	$\delta < 0.1$	$\delta < 0.03$	SR (%)	TT (sec)	$\delta < 0.5$	$\delta < 0.1$	$\delta < 0.03$
L&F [42]	60.00	15.80	75.11	70.10	66.04	60.00	17.12	81.44	66.70	62.86	60.00	13.35	63.76	55.59	55.59
MPC [11]	100.00	20.10	70.43	69.17	66.33	100.00	14.79	80.61	72.89	67.85	100.00	18.21	78.24	70.48	63.62
DGNN [43]	100.00	34.70	60.41	58.91	58.91	100.00	63.70	47.85	42.33	41.92	100.00	36.80	27.90	20.16	20.16
AFOR [40]	100.00	30.50	92.89	90.40	88.66	100.00	51.50	91.39	90.35	87.96	100.00	183.50	88.60	85.13	72.90
STAF [44]	100.00	84.80	81.56	71.69	70.59	100.00	58.80	77.22	71.79	69.80	100.00	90.06	91.13	79.53	77.43
Diff.	100.00	19.32	65.73	56.04	55.26	100.00	20.89	56.81	54.46	52.07	100.00	15.22	59.12	55.24	51.41
ReDiG (ours)	100.00	13.10	84.51	81.52	78.97	100.00	12.46	82.43	81.21	80.37	100.00	10.03	91.17	87.68	80.91

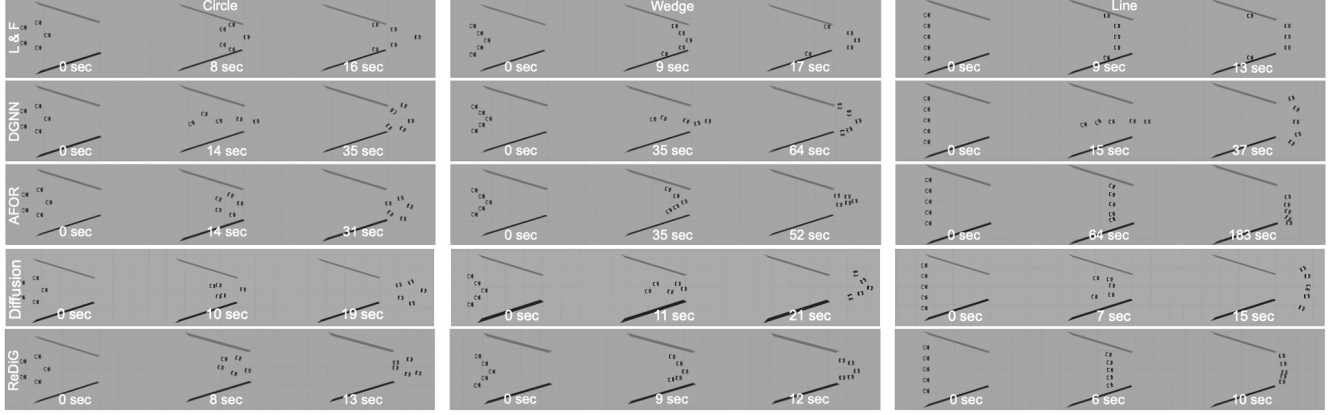


Fig. 3. Qualitative results from Gazebo simulations with Limo robots, illustrating formation adaptation across three distinct formations.

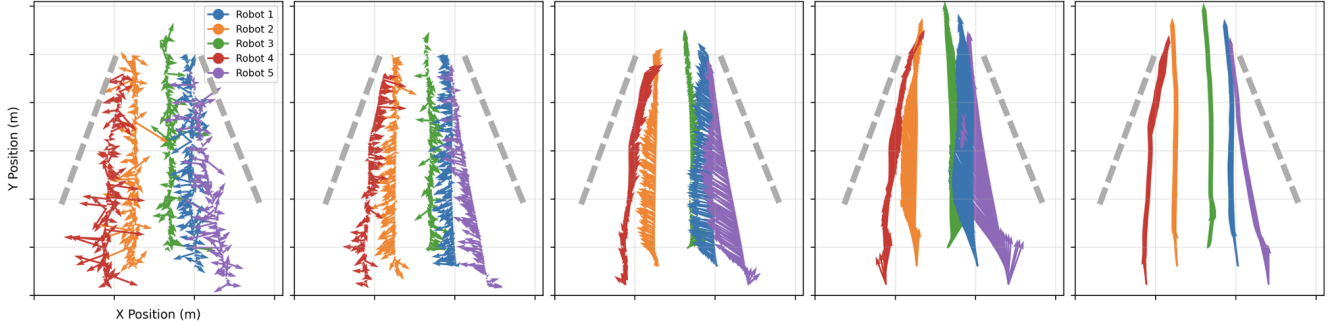


Fig. 4. Visualization of ReDiG denoising (left to right) for learning navigation policies in a wedge-formation robot team. Colored arrows denote each robot's velocity at a diffusion step; dashed gray lines indicate obstacles.

V. EXPERIMENTS

A. Experimental Setups

We comprehensively evaluate our ReDiG approach in three experimental settings: (1) a Gazebo simulation using ROS2, (2) a high-fidelity Unity-based 3D multi-robot simulator in ROS1, and (3) a physical robot team running ROS2. Each setup involves different differential-drive robot platforms (e.g., Limo and Warthog robots), and formation shapes (e.g., circle, wedge, and line). To follow the identical trajectories, we convert the linear velocity $\mathbf{a}_i$ into corresponding wheel velocities. All scenarios feature narrow corridors, where the robot teams are required to navigate through confined spaces while dynamically adjusting their formation to preserve its overall structure. In simulation, robot positions and environmental obstacles are obtained directly from the Gazebo and Unity simulation. For physical experiments, each robot performs state estimation and environment mapping using a SLAM method [45].

To demonstrate the effectiveness of ReDiG, we compare

it with five prior methods and one baseline, including: (1) *Leader and Follower (L&F)* [42], where a leader guides the team while followers maintain formation; (2) *Model Predictive Control (MPC)*, an optimization-based method that encodes formation control as a cost function; (3) *Decentralized GNN (DGNN)* [43], a GNN-based method for decentralization without formation control; (4) *Adaptive Formation with Oscillation Reduction (AFOR)* [40], an online RL approach for formation adaptation; (5) *SubTeaming and Adaptive Formation (STAF)* [44], which enables splitting with adaptive formation; and (6) **Diff.**, a diffusion-only baseline without online RL refinement.

To quantitatively evaluate and compare ReDiG with other methods, we use three metrics, including: (1) *Success Rate (SR)* measures the proportion of robots in the team that reach their goal without collisions; (2) *Travel Time (TT)* represents the total team navigation time; (3) *Contextual Formation Integrity (CFI)* [23] measures the real-time adherence of the robots to their designated formation, based on a threshold δ that determines how strictly the formation shape must be preserved.

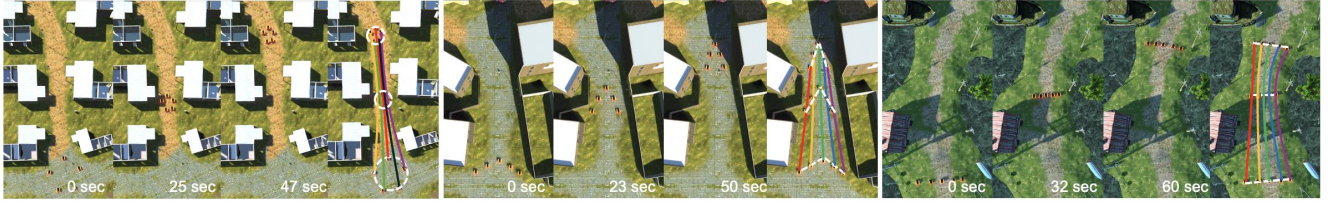


Fig. 5. Qualitative results from Unity3D simulations with differential-drive robots, demonstrating formation adaptation while navigating long, narrow corridors.

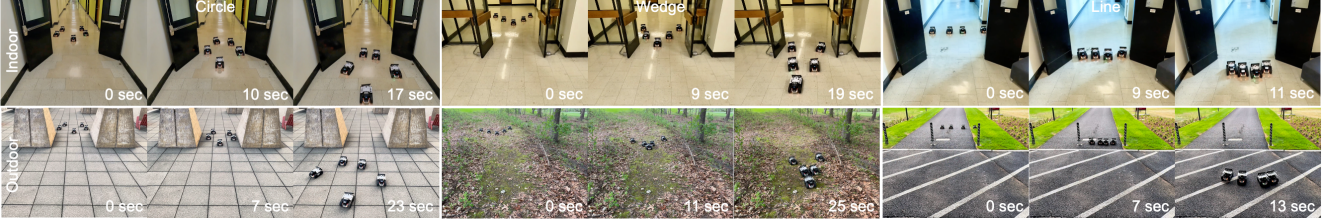


Fig. 6. Qualitative results from real-world environments using decentralized teams of Limo robots running ROS2, and communicating via Wi-Fi broadcasting.

B. Results in Multi-Robot Simulations

The qualitative results from Gazebo simulations are presented in Figure 3. The L&F method, which relies on a predefined rigid formation, fails to navigate narrow corridors, as the outer robots collide with the walls. DGNN does not incorporate formation control, resulting in robots passing through the corridor sequentially without coordination. Diffusion removes the online RL refinement from ReDiG and thus cannot improve its policy through reward feedback to enable formation adaptation; it only supports individual robot navigation. Both AFOR and our proposed ReDiG approach integrate a spring-damper model to enable formation adaptation. However, AFOR is built upon RL with discrete action decisions, which result in jerky trajectories. In contrast, ReDiG generates significantly smoother trajectories by leveraging a diffusion-based policy.

We visualize the denoising process of ReDiG for a robot team in wedge formation, as shown in Figure 4. Each arrow represents a velocity vector generated at different denoising steps. Starting from pure Gaussian noise, the actions of each robot are progressively denoised into smooth, coordinated motion. This demonstrates ReDiG’s ability enable both smooth navigation and formation adaptation.

The quantitative results are shown in Table I. The L&F method achieves a 60% success rate due to its inability to adapt formations. DGNN, which lacks formation awareness, performs worst in the CFI metric. AFOR and STAF yield longer travel times because sequential multi-robot RL often produces jerky, inefficient motion. Diffusion attains shorter travel time, but its CFI degrades since it cannot refine the policy to support formation adaptation. ReDiG addresses these limitations and achieves above 82% CFI and the shortest travel time across all three formation shapes. ReDiG has slightly lower CFI scores compared to AFOR, which is a reasonable trade-off for the significant gain in efficiency. These results highlight the effectiveness of ReDiG in enabling smooth and efficient formation adaptation in complex environments.

We compare ReDiG with the baseline methods in terms of motion smoothness. We use the team-average jerk metric [46], defined as $\mathbf{j}(t) = \frac{1}{n} \sum_{i=1}^n \frac{d^3 \mathbf{p}_i(t)}{dt^3}$, where n is the number of robots and lower values indicate smoother motion. The quantitative results are shown in Table II. DGNN, AFOR, and STAF exhibit relatively higher jerk, as their RL-based controllers directly generate actions with discrete decision-making.

MPC achieves lower jerk due to its short-horizon optimization over control sequences, which promotes more temporally consistent motion. ReDiG achieves the lowest mean jerk of 0.080, outperforming all baselines and reducing jerk by 40.3% compared with Diff. This improvement indicates that online RL refinement with the spring-damper reward further enhances motion smoothness. Overall, the results demonstrate that ReDiG’s diffusion policy enables smooth and coordinated robot trajectories.

Beyond Gazebo, we evaluate our approach in a Unity3D-based simulator integrated with ROS1 for multi-robot perception and control. These outdoor environments include extended narrow pathways between buildings and uneven flooded terrain, requiring long, curved trajectories with adaptive formation control. As shown in Figure 5, ReDiG enables Warthog teams to adapt their formation to environmental constraints, navigating smoothly and reaching goals without collisions. In circle formation scenarios, the team traverses narrow corridors by continuously reshaping the formation for tighter spaces. These results highlight ReDiG’s effectiveness in achieving smooth, adaptive formation control in complex environments.

C. Validation on Physical Robot Teams

We validate ReDiG through case studies using physical differential-drive Limo robots, each equipped with an onboard Intel NCU i7 processor for real-time execution. The robots

TABLE II
COMPARISON OF
MEAN JERK.

Method	Mean Jerk ↓
L&F [42]	0.251
MPC [11]	0.152
DGNN [43]	0.443
AFOR [40]	0.171
STAF [44]	0.165
Diff.	0.134
ReDiG (ours)	0.080

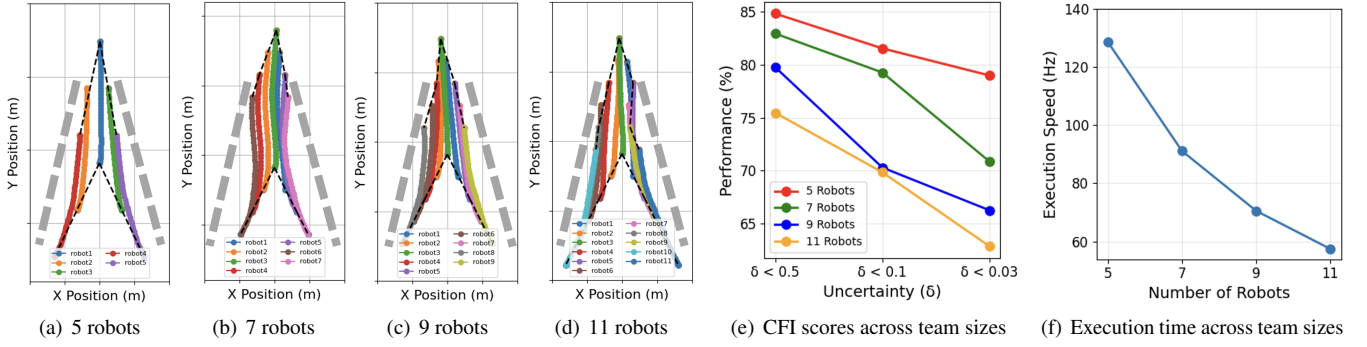


Fig. 7. Quantitative results of ReDiG scalability. Figures (a)-(d) illustrate trajectories of teams with 5-11 robots performing formation adaptation in a wedge formation. Figure (e) reports the CFI across varying team sizes and δ values, and Figure (f) shows the execution time as a function of team size.

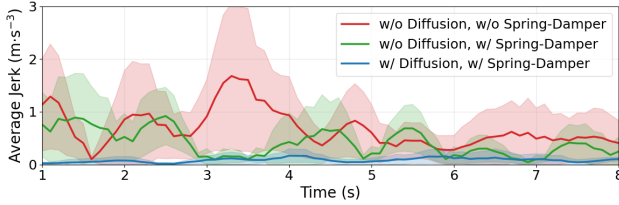


Fig. 8. Ablation study on motion trajectory smoothness.

run ROS2 and coordinate via Wi-Fi-based broadcasting for decentralized communication. Experiments were conducted in both indoor and outdoor environments. As shown in Figure 6, our method enables teams of varying sizes to smoothly adapt formation while navigating narrow indoor spaces. The outdoor experiments include narrow passages bordered by bollards, scattered trees, and roadblocks. Notably, in wedge formation scenarios, ReDiG effectively guided the team through uneven forest terrain where wheel slippage introduced movement uncertainty. These results further validate the effectiveness of our approach and its applicability to real-world coordinated navigation with formation adaptation.

D. Discussion

Scalability: We evaluate ReDiG’s scalability with respect to team size. Figures 7(a)-7(d) show qualitative results for 5-11 robots, where the team successfully traverses narrow corridors with formation adaptation. As shown in Figure 7(e), ReDiG maintains around 80% CFI for 5 robots at $\delta < 0.3$, and achieves about 75% for 11 robots at $\delta < 0.5$. Figure 7(f) further reports execution time across team sizes; maintains a real-time control frequency of 57.4 Hz for 11 robots. These results align with the execution-time complexity of $\mathcal{O}(n + T_d)$, validating ReDiG’s scalability from both theoretical and experimental perspectives for decentralized coordinated navigation with smooth formation adaptation in larger robot teams.

Ablation Study on Motion Trajectory Smoothness: We conduct an ablation study to evaluate the role of the diffusion and the spring-damper model in enhancing motion trajectory smoothness. Figure 8 shows the jerk profiles over time for a team of five robots. The red line shows the case without diffusion and without the spring-damper model, where the jerk is the highest. The green line corresponds to using the spring-damper model without diffusion; the damping effect reduces jerk, but noticeable fluctuations remain. The blue line

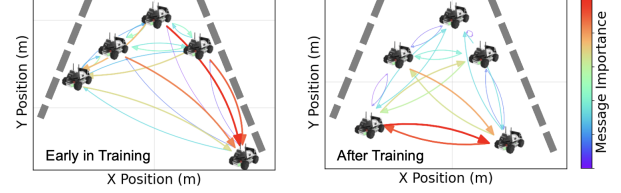


Fig. 9. Comparison of inter-robot message importance across training stages.

combines both diffusion and the spring-damper model, where iterative refinement of robot actions achieves the lowest jerk. These results highlight the effectiveness of the diffusion model in producing smooth individual robot trajectories.

Ablation Study of Message Passing Informativeness: To understand how graph learning facilitates team coordination, we conduct a message importance study. The graph network uses learnable weights $\mathbf{W}^h$ to compute team embeddings $\mathbf{h}_i$ through message passing. We quantify the importance of messages exchanged from robot j to i as $\|\mathbf{W}^h(\mathbf{z}_j - \mathbf{z}_i)\|_2$ over training, as shown in Figure 9. Early in training, the right-most robot distributes attention uniformly across teammates, failing to capture team structure and resulting in a wall collision. After training, the outermost robots assign the highest importance to each other, reflecting their critical role in regulating team size. These results highlight the effectiveness of graph learning for coordinated formation adaptation.

VI. CONCLUSION

We propose ReDiG as a principled approach for decentralized coordinated navigation with smooth formation adaptation. ReDiG leverages graph-conditioned diffusion policies refined via online actor-critic reinforcement learning, with a spring-damper model to enable smooth, adaptive formation control in a unified learning framework. Extensive experiments demonstrate that ReDiG achieves smooth formation adaptation and state-of-the-art performance in coordinated multi-robot navigation. ReDiG exhibits several limitations. First, it requires retraining when switching between different formation types. Second, it requires single-robot trajectories as initialization to improve sample efficiency. Third, its implementation is evaluated on a 2D surface. Future work will address these limitations through formation-conditioned policies, self-supervised learning to reduce initialization dependence, and extension to 3D environments with terrain-adaptive formation control.

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